

Title: Laser Security Alarm using LDR and Transistor.

Objectives:

This project demonstrates a simple laser-based security alarm using a **Light Dependent Resistor (LDR)** and a **Transistor switch**. The system works by pointing a laser beam continuously at the LDR. Under normal conditions, the laser keeps the resistance of the LDR low. When the beam is interrupted (e.g., someone walks through it), the LDR's resistance increases, which affects the voltage across a voltage divider and turns **on a transistor switch**. The transistor then activates a **buzzer** (and optionally an **LED**) as an alert system.

This project utilizes:

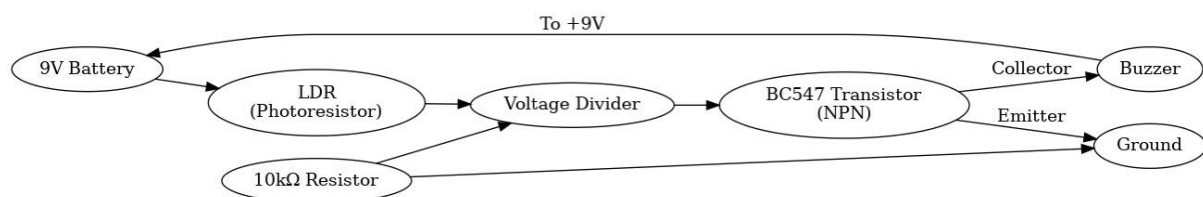
- Photosensitivity of the LDR for detecting light.
- Voltage divider rule for analog signal control.
- Transistor switching for digital control of output devices like buzzers.

This circuit models how many modern intruders alarm systems work, especially in laser tripwire systems used in banks, museums, and restricted zones.

Theory and Methodology:

An LDR is a variable resistor whose resistance decreases when exposed to light and increases in darkness. This property is used to detect the presence or absence of a laser beam. When the LDR is exposed to the laser, its resistance is low. When the laser is interrupted (by someone walking through it, for example), the resistance rises significantly.

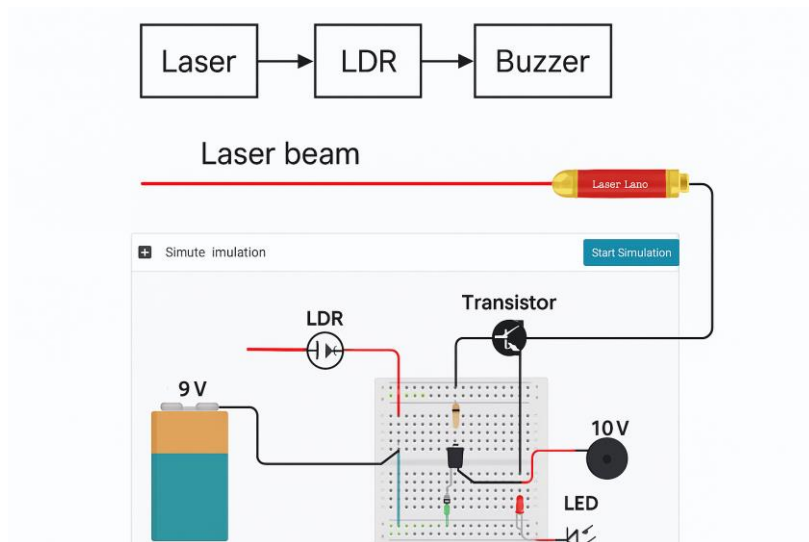
A voltage divider circuit is built using the LDR and a fixed resistor ($10\text{k}\Omega$), which generates a varying voltage output based on the LDR's resistance. This voltage is fed into the base of a BC547 NPN transistor.



The transistor acts as a switch. When the base voltage is high enough (typically above 0.7V), the transistor remains off, and the buzzer stays silent. When the base voltage drops (due to increased LDR resistance in darkness), the transistor turns on, allowing current to flow from collector to emitter, activating the buzzer or LED as an alarm.

Key Concepts:

- **LDR:** A light-sensitive resistor that changes resistance based on light intensity.
- **Voltage Divider:** A circuit that uses two resistors (LDR + fixed resistor) to divide voltage.
- **Transistor (BC547):** Acts as an electronic switch.
- **Buzzer:** Gives an audible alarm when triggered.

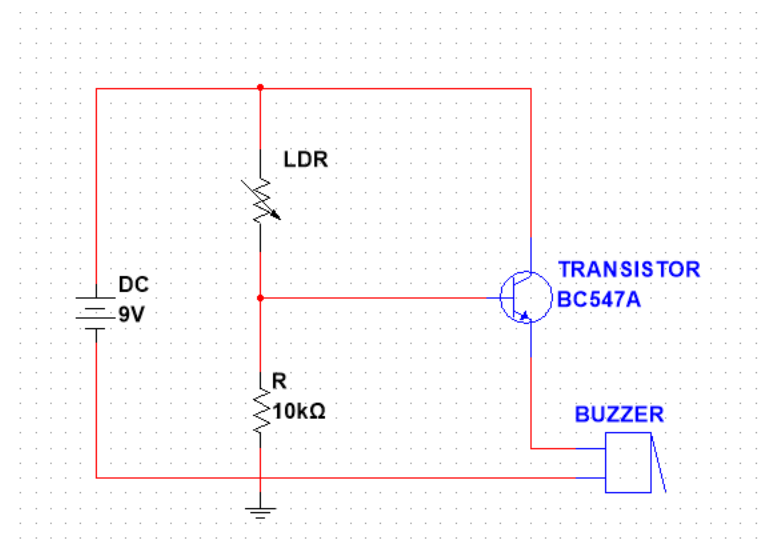


Process of Laser Security Alarm

Equipment:

1. 9V DC Battery
2. Battery Clip
3. Breadboard
4. Laser Light
5. LDR (Light Dependent Resistor)
6. $10k\Omega$ Resistor
7. BC547 NPN Transistor
8. Buzzer
9. LED (Optional for visual alert)
10. Connecting Wires

Circuit Diagram:



Circuit Diagram for Laser Security Alarm

Procedure:

- 1) 9V battery had been connected to the mini breadboard using a battery clip.
- 2) LDR and 10kΩ resistor had been inserted in a voltage divider configuration.
 - One end of the LDR had gone to the positive terminal of the battery.
 - The other end had been connected to one side of the 10kΩ resistor.
 - The other end of the 10kΩ resistor had been connected to the ground (GND).
 - The junction between the LDR and resistor had been connected to the base of the BC547 transistor.
- 3) Emitters of the transistor had been connected to GND.
- 4) Collector of the transistor had been connected to one end of the buzzer.
 - The other end of the buzzer had gone to the +9V.
- 5) Laser light had been pointed directly at the LDR so that it received continuous light.
- 6) The laser beam had been interrupted, and the buzzer had activated.

Data and Analysis:

LDR Resistance under Conditions

Condition	LDR Resistance	V _{Base} (Approx)	Transistor State	Buzzer
Laser on LDR	~ 1kΩ	~ 8.2V	OFF	Silent
Laser interrupted	~ 100kΩ	~ 0.82V	ON	ON

Calculation,

Voltage Divider Formula,

$$V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$$

Here,

$$R_1 = \text{LDR}$$

$$R_2 = 10\text{k}\Omega$$

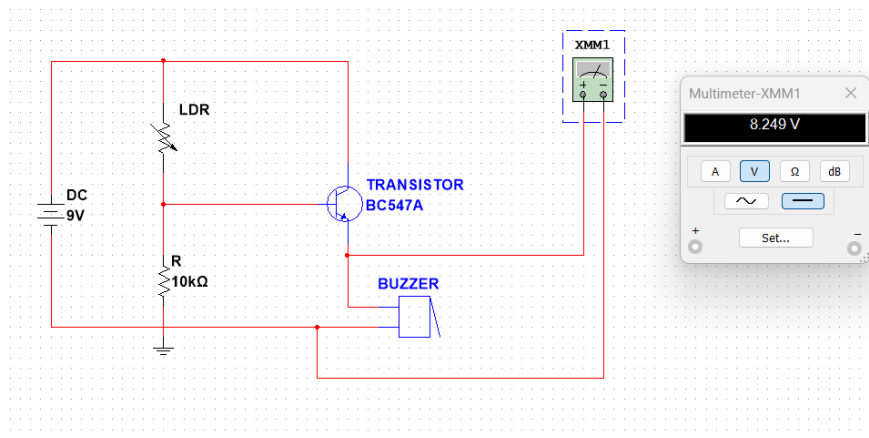
$$V_{in} = 9\text{V}$$

Example Calculation (Laser Block),

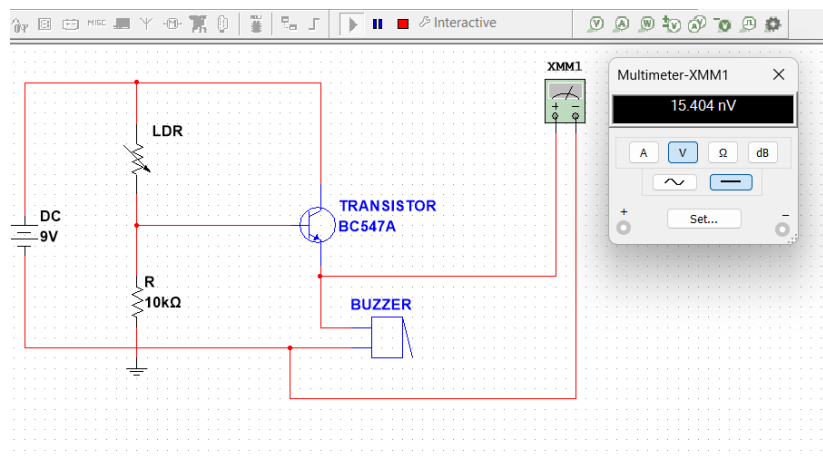
$$V_{out} = 9 \times \frac{10\text{k}}{100\text{k} + 10\text{k}} = 9 \times 0.090982\text{V}$$

$$= 0.7\text{ V}$$

When $V_{base} < 0.7\text{V}$, the transistor turns ON, and the buzzer Sounds.

Simulation:

LDR is exposed to the laser, its resistance is low



LDR's Resistance increases $\Rightarrow$ Turn on Transistor Switch $\Rightarrow$ Buzzer Active

Discussion:

This experiment demonstrates how light interruption can be used for intrusion detection, highlighting the interplay between analog and digital components. The LDR and resistor create an analog response to changes in light intensity, while the transistor acts digitally, switching fully on or off depending on the input voltage at its base.

In our test,

- When the laser beam was uninterrupted, the LDR received constant light, its resistance stayed low, and the voltage at the transistor's base remained high enough to keep the transistor off, so the buzzer was silent.
- When the laser beam was blocked, the LDR entered darkness, increasing its resistance. This dropped the base voltage below the switching threshold of the transistor ($\sim 0.7V$), causing the transistor to turn on and activate the buzzer/LED.

This circuit is highly sensitive to light levels, making it suitable for security but also prone to false triggering due to ambient light changes or misalignment of the laser. Improvements such as shielding the LDR or adding a comparator IC could improve reliability.

Applications and Extensions:

- Used in home and office security systems.
- It can be integrated with microcontrollers (e.g., Arduino) for smart monitoring.
- Multiple LDR-laser setups can cover larger areas.

Limitations:

- Sensitive to environmental light fluctuations.
- It depends on the proper laser alignment.
- Not suitable for outdoor use without weatherproofing.

Conclusion:

This project successfully demonstrates a basic laser-based security alarm system using an LDR, transistor, and buzzer. The system relies on changes in light intensity to trigger an alert. It is an effective example of how light-sensitive circuits and transistor switching work in real-world security applications. This project enhances understanding of voltage dividers, transistor operation, and sensor-based automation.